

MOCCM Quant Hackathon 2026

The Black-Box Signal Hunt

Mathematical Strategy Proof

Team Blackjacks

Arya Patil Mudil Goel Rutuparn Ranade

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Results Summary

Metric	Long-Only	Long-Short
Net Sharpe Ratio	10.4598	10.8957
Blended Sharpe	10.6778	
Max Gross Exposure	\$100,000	\$200,000
Starting Cash	\$1,000,000	\$2,000,000
Grader Status	PASS	PASS
Runtime	7.16 seconds	

1 Executive Summary

We discovered a deterministic, multi-dimensional, white-box trading signal embedded in the MOCCM synthetic intraday dataset. The signal is a **four-ticker conditional momentum strategy**: TICKER_01's return at bar t predicts TICKER_00's directional movement, but only when two regime conditions - defined by TICKER_02 volume and TICKER_38 return - are simultaneously satisfied, and when TICKER_00 itself is sufficiently active.

The strategy is **fully white-box**: every condition is an arithmetic threshold that can be written as an **if**-statement and defended mathematically. No machine learning, date filters, or black-box models are used. The final rule produces a **Blended Net Sharpe Ratio of 10.6778** on the official MOCCM grader, validated across the full 10-year training dataset as well as first and second 5-year halves independently.

Window	Long-Only Sharpe	Long-Short Sharpe
Full 10-year	10.4598	10.8957
First 5-year half	11.1367	11.4866
Second 5-year half	9.7608	10.7278
Quarter 1	10.4300	10.6642
Quarter 2	12.1408	12.7326
Quarter 3	9.9061	11.1959
Quarter 4	9.8891	10.4717

The signal is stable across every sub-period, which is the strongest available evidence that the relationship is structural rather than a coincidental artifact of a specific era in the synthetic data.

2 Signal Discovery Methodology

2.1 Why Returns and Volume?

The dataset provides exactly two raw columns per ticker: **Close** and **Volume**. From these, the most natural and defensible white-box features are:

- **Return:** $r_X(t) = \text{Close}_X(t)/\text{Close}_X(t-1) - 1$. Scale-free and comparable across tickers regardless of price level.
- **Absolute return:** $|r_X(t)|$. Measures activity or volatility regardless of direction. Useful when direction is supplied by another ticker.
- **Volume:** $V_X(t)$. Measures market activity and often encodes regime switches in synthetic datasets.

Rolling sums, raw price levels, and time-of-day filters were all considered and rejected. Rolling features did not improve after-fee Sharpe. Raw price levels are non-stationary and harder to generalise. Time filters would constitute memorisation of the training dataset rather than discovery of a structural signal.

2.2 Step 1 - Lead-Lag Scan: Why TICKER_01?

The discovery process began with a systematic one-step lead-lag correlation scan. For each predictor ticker $X \in \{\text{TICKER_01}, \dots, \text{TICKER_49}\}$, we computed:

$$\rho_X = \text{Corr}(r_{00}(t), r_X(t-1)), \quad t = 1, \dots, T$$

The result was unambiguous. **TICKER_01** produced a lag-1 Pearson correlation of $\rho = 0.2277$ with **TICKER_00**. The next strongest predictor reached only 0.005 - roughly 45 times weaker. This is not a marginal statistical difference; it is a structural anomaly that immediately identifies **TICKER_01** as the primary directional signal.

Ticker	Lag-1 Corr with TICKER_00	Classification
TICKER_01	0.2277	Primary signal
TICKER_28	0.0050	Noise
TICKER_47	0.0046	Noise
TICKER_42	0.0039	Noise
All others	< 0.005	Noise

To confirm the relationship is linear and directional (not just correlated in magnitude), we binned **TICKER_01** lag-1 returns into deciles and computed the average **TICKER_00** return per decile:

Decile	0	1	2	3	4	5	6	7	8	9
Avg r_{00} (%)	-0.148	-0.083	-0.057	-0.031	-0.013	+0.012	+0.035	+0.059	+0.089	+0.149

The relationship is **perfectly monotonic**: every decile is higher than the previous one, with no exceptions. This proves the directional relationship is genuine and not driven by outliers or a particular regime.

We further verified that the same-bar return $r_{01}(t)$ and the lag-1 return $r_{01}(t-1)$ predict **TICKER_00**'s next bar identically:

Same-bar signal, next-bar return:	+0.2507%
Lagged signal, next-bar return:	+0.2507%

confirming that the competition's Market-On-Close convention makes same-bar features legitimate. The same information is available either way; using the same-bar return is not lookahead.

2.3 Step 2 - Volume Conditioning: Why TICKER_02?

After identifying TICKER_01 as the direction source, we asked: *when* is this signal reliable? We tested volume and return conditions across all remaining tickers. Filtering to bars where TICKER_02 volume exceeded its median (1,980 units) raised the conditional correlation from 0.2277 to 0.402.

The interpretation is natural: TICKER_02 volume acts as a **market-state switch**. When TICKER_02 is active (high volume), the cross-asset lead-lag relationship between TICKER_01 and TICKER_00 strengthens materially. When volume is low, the relationship is weaker and 10 bps friction tends to dominate any edge.

We tested alternative volume gates (other tickers, different thresholds). No other single ticker's volume matched TICKER_02 as a regime discriminator: the best alternatives produced blended Sharpe around 8.0, compared to the 10.68 achieved with TICKER_02.

2.4 Step 3 - Target Self-Confirmation: Why TICKER_00 Itself?

Using TICKER_00's own absolute return as a filter is not circular. The direction signal comes entirely from TICKER_01. The role of $|r_{00}(t)|$ is purely to confirm that the target is active enough to trade.

The economic logic is simple: if TICKER_00 barely moved in the current bar, the expected continuation is tiny and the 10 bps round-trip fee will dominate. If the target is already moving ($|r_{00}(t)|$ above threshold), the lead-lag edge is large enough to be profitable after costs. Absolute return (not signed return) is used because both upward and downward active bars can carry continuation/lead-lag information.

Thresholds were chosen to balance quality (higher threshold = better signal quality) against quantity (higher threshold = fewer events = less stable Sharpe): $|r_{00}(t)| \geq 0.0036$ for Long-Only and ≥ 0.0034 for Long-Short.

2.5 Step 4 - Fourth-Ticker Search: Why TICKER_38?

The organiser hint confirmed four tickers are involved including TICKER_00. With three tickers fixed, we scanned all remaining tickers as the fourth regime filter, testing return filters, absolute return filters, and volume filters:

Candidate	Best Logic	Approx. Blended Sharpe	Decision
No fourth ticker	Only 00/01/02	9.40	Useful baseline
TICKER_37	Volume < 5,000	9.03-9.29	Rejected
TICKER_16	Return ≥ -0.00322	9.44	Rejected
TICKER_38	Return < 0.0050 (LO)	10.68	Selected

TICKER_38 acts as a **noise-state filter**. Its role is not directional: the condition is that its current return stays below a small positive cap. When TICKER_38 has an extreme positive move, the TICKER_01 $\rightarrow$ TICKER_00 relationship weakens. Filtering out these bars removes noisy trades and materially improves after-fee Sharpe.

3 Mathematical Signal Definition

3.1 Notation

$$r_X(t) = \frac{\text{Close}_X(t)}{\text{Close}_X(t-1)} - 1 \quad (\text{5-minute return of ticker } X \text{ at bar } t)$$

$$V_{02}(t) = \text{Volume of TICKER.02 at bar } t \quad P_0(t) = \text{Close price of TICKER.00 at bar } t$$

$$T = \text{total number of 5-minute bars}$$

3.2 Active Regime

Long-Only active regime $R^{\text{LO}}(t) = 1$ if and only if *all* of:

$$\begin{cases} V_{02}(t) \geq 2500 & (\text{liquidity gate}) \\ r_{38}(t) < 0.0050 & (\text{volatility noise filter}) \\ |r_{00}(t)| \geq 0.0036 & (\text{target activity filter}) \\ r_{01}(t) \text{ is finite} & (\text{data quality guard}) \end{cases}$$

Long-Short active regime $R^{\text{LS}}(t) = 1$ if and only if *all* of:

$$\begin{cases} V_{02}(t) \geq 2500 \\ r_{38}(t) < 0.00425 \\ |r_{00}(t)| \geq 0.0034 \\ r_{01}(t) \text{ is finite} \end{cases}$$

3.3 Trade Signal

Long-Only:

$$S^{\text{LO}}(t) = \begin{cases} +1 & \text{if } R^{\text{LO}}(t) = 1 \text{ and } r_{01}(t) \geq +0.0016 \\ 0 & \text{otherwise} \end{cases}$$

Long-Short:

$$S^{\text{LS}}(t) = \begin{cases} +1 & \text{if } R^{\text{LS}}(t) = 1 \text{ and } r_{01}(t) \geq +0.0026 \\ -1 & \text{if } R^{\text{LS}}(t) = 1 \text{ and } r_{01}(t) \leq -0.0026 \\ 0 & \text{otherwise} \end{cases}$$

The Long-Short threshold (0.0026) is stricter than Long-Only (0.0016) because the strategy must be symmetric: weak short signals generate negative returns and must be excluded alongside weak long signals.

3.4 Position Sizing

$$\text{target_exposure}(t) = \min(\text{exposure_cap}, \max(0, \text{NAV}(t)) \times \text{fraction})$$

$$\text{target_shares}(t) = \frac{S(t) \times \text{target_exposure}(t)}{P_0(t)}$$

Strategy	Direction	Fraction	Abs. Cap
Long-Only	Long	10% of \$1M cap	\$100,000
Long-Short	Long	10% of \$2M cap	\$200,000
Long-Short	Short	60% of \$2M cap	\$200,000

The 10% fraction is deliberately conservative. It keeps gross exposure far below the cap, makes margin violations nearly impossible, and ensures that 10 bps per trade is a small percentage of total capital, preventing fee drag from wiping out equity.

3.5 Core Accounting Identities

$$\Delta \text{shares}(t) = \text{target_shares}(t) - \text{shares}(t-1) \quad (1)$$

$$\text{Interval_Turnover}(t) = |\Delta \text{shares}(t)| \times P_0(t) \quad (2)$$

$$\text{Cash}(t) = \text{Cash}(t-1) - \Delta \text{shares}(t) \times P_0(t) \quad (3)$$

$$\text{Gross_Exposure}(t) = |\text{shares}(t) \times P_0(t)| \quad (4)$$

$$\text{Gross_NAV}(t) = \text{Cash}(t) + \text{shares}(t) \times P_0(t) \quad (5)$$

$$\text{Net_NAV}(t) = \text{Gross_NAV}(t) - \sum_{\tau=1}^t \text{Turnover}(\tau) \times 0.001 \quad (6)$$

4 No Lookahead Bias - Formal Proof

The MOCCM rulebook defines lookahead as “ingestion of forward data points.” Since the dataset contains only a single **Close** price and **Volume** per 5-minute bar (no Open, High, or Low), the only executable price at bar t is $\text{Close}_{00}(t)$. This is the Market-On-Close (MOC) convention: all interval- t observations are available simultaneously at the close of bar t , and trades execute at that same close.

Under this convention, every piece of data used in our signal is from bar t or earlier:

Data item	Bar	Available at	Lookahead?
$r_{01}(t)$	t	End of bar t	No
$V_{02}(t)$	t	End of bar t	No
$r_{38}(t)$	t	End of bar t	No
$r_{00}(t)$	t	End of bar t	No
$P_0(t) = \text{Close}_{00}(t)$	t	End of bar t	No (execution price)
$r_{01}(t+1)$	$t+1$	Future	Never used

The engine enforces the guard condition $0 < i < n-1$, preventing any signal from firing on the first or last bar. The first and last **Gross_Exposure** values are exactly zero by construction.

5 Portfolio Constraint Management

Capital ceilings. At every bar, position size is computed as $\min(\text{exposure_cap}, \text{NAV} \times \text{fraction})$. The exposure cap is \$100,000 (Long-Only) and \$200,000 (Long-Short), representing 10% of the respective capital limits. A secondary price-drift guard trims positions if mark-to-market value exceeds the cap between trades.

Transaction fee survival. The 10 bps fee applies to every dollar of `Interval_Turnover`. With a maximum position of \$100,000–\$200,000, the cost per trade is at most \$100–\$200. The four-condition regime gate keeps trade frequency selective, making cumulative fees negligible relative to gross returns.

Flat start and end. Signal arrays are forced to zero at indices 0 and $T - 1$ before the backtest loop. A forced-liquidation block at the final bar closes any remaining open position, ensuring $\text{Gross_Exposure}[0] = \text{Gross_Exposure}[T - 1] = 0$.

Margin safety. Positions are sized as $\min(|\text{Cash}|, \text{cap})$: the strategy never deploys more than available cash. Short entries credit proceeds to cash, keeping the balance non-negative throughout.

Dual portfolio construction.

- **Long-Only:** Acts only on positive `TICKER_01` signals; stays in cash when the signal is absent or negative. Protection against drawdowns comes from selectivity and low exposure, not from shorting.
- **Long-Short:** Trades both sides of the `TICKER_01` signal. The short leg is not reckless leverage; it uses the same 10% fraction on a \$2M account, capped at \$200,000. The result is genuine directional diversification: negative `TICKER_01` states generate profit on the short side instead of being ignored.

6 Systems Architecture and Engineering

Technology stack. Python 3 with NumPy and pandas (only for reading csv). No machine learning libraries are imported. The code is deterministic and inspectable.

Speed optimisations.

1. `usecols` filter on `pd.read_csv`: reads only `Timestamp`, `Ticker`, `Close`, `Volume`.
2. Ticker pre-filter before pivot: retains only the 4 required tickers out of 50, reducing the pivot matrix by $\approx 92\%$.
3. Fully vectorised return and volume arrays via NumPy.
4. Sequential backtest loop: $O(T)$ over timestamps; no nested loops.

Runtime on the 10-year training file (189,000 timestamps): **7.16 seconds**. The hidden 5-year file ($\approx 94,500$ timestamps) will run in approximately 3.5 seconds.

Grader invariants verified locally.

$$|\text{Gross_NAV}(t) - \text{Cash}(t)| = \text{Gross_Exposure}(t) \quad \forall t$$

$$|\Delta \text{Cash}(t)| = \text{Interval_Turnover}(t) \quad \forall t \geq 1$$

$$\text{Gross_Exposure}(t) \leq \text{cap} \quad \forall t \quad \text{Cash}(t) \geq 0 \quad \forall t$$

7 Validation and Grader Output

7.1 Official Grader Output

```
Evaluating Team : BLACKJACKS
Long-Only      : PASS | Pass
Long-Short     : PASS | Pass
BLACKJACKS , Blended_Sharpe = 10.6778 , Status = Valid
```

7.2 Is This Curve Fitting?

The honest answer is: parameter-fit risk exists, but this is not black-box overfitting.

- **Why it is not pure curve fitting:** Only four tickers are used. Every condition has an interpretable economic role. No date filters, time-of-day filters, ticker-specific weights, or learned classifiers are present. The rule is stable across both 5-year halves and all four quarters with no sign reversal.
- Since the signal is “deterministic,” both train and test datasets are generated by the same synthetic process, a correctly identified deterministic rule will generalise perfectly.

8 Conclusion

We isolated the hidden signal as a **four-ticker conditional momentum rule**. TICKER_01's return at bar t predicts TICKER_00's directional movement, gated by TICKER_02's volume (liquidity regime), TICKER_38's return (noise-state filter), and TICKER_00's own absolute return (target activity confirmation). The rule is fully white-box, mathematically explicit, lookahead-free, and achieves a **Blended Net Sharpe Ratio of 10.6778** on the official grader - stable across every time period tested.
